

PROJECT WORK CS6811
SECOND REVIEW

Federated Learning for Code Generation with Hallucination-Aware Fine Tuning

TEAM DETAILS

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DOMAIN: Large Language Models

INTRODUCTION

Large Language Models (LLMs) have remarkable capabilities in code generation, enabling automated solutions to a wide range of software engineering tasks. However, LLMs frequently suffer from hallucinations in code generation, producing syntactically valid but semantically incorrect programs, non-existent APIs, or logically flawed implementations. Existing approaches to mitigating code hallucinations primarily rely on prompt engineering, repeated sampling, or full model fine-tuning. Full fine-tuning is computationally expensive and impractical in low-resource or decentralized settings, and repeated sampling does not address the underlying causes and mitigation of hallucination. In real-world development environments, code generation systems are often deployed across multiple decentralized clients where data cannot be centrally aggregated due to privacy, ownership.

To address these limitations we propose a hallucination-aware code generation framework that combines program analysis, targeted automatic program repair, and federated parameter-efficient fine-tuning using LoRA

PROBLEM STATEMENT

Code generation using large language models (LLMs) is currently hindered by hallucinations. Existing mitigation strategies typically rely on supervised fine-tuning or reinforcement learning from human feedback (RLHF), where models are trained to align with human preferences. While parameter-efficient fine-tuning methods such as LoRA reduce computational overhead compared to full fine-tuning, both approaches primarily optimize model behavior but without explicit verification of program correctness. As a result, these methods may improve correctness but often fail to systematically detect and correct deeper structural, semantic, and execution-level errors in generated code.

Furthermore, in real world scenarios, high-quality code and execution feedback are distributed across multiple organizations making centralized data collection infeasible due to privacy concerns. Without a secure and decentralized learning mechanism, models are unable to effectively learn from diverse hallucination patterns observed across clients. This creates a critical gap for a unified framework that integrates explicit hallucination detection and automated program verification with privacy-preserving, parameter-efficient adaptation to improve the quality of code generation.

OBJECTIVES

- To design a hallucination-aware code generation framework that integrates static and dynamic program analysis.
- To detect and classify different types of hallucinations in generated code.
- To apply automatic program repair for correcting hallucinated code using knowledge-guided patching.
- To fine-tune large language models using parameter-efficient LoRA in a federated learning setting.
- To evaluate the proposed framework on standard code generation benchmarks and analyze hallucination reduction.

LITERATURE SURVEY

S.NO	TITLE	NAME OF THE JOURNAL	METHODOLOGY	LIMITATIONS
[1]	Advancing LLM-Generated Code Reliability: A Hybrid Approach for Hallucination Detection	IEEE Transactions on Software Engineering (TSE), 2025	• Proposes SDHD, a hybrid hallucination detection framework that integrates static analysis and dynamic execution for LLM-generated code. • This framework detects eight hallucination types and achieves superior precision, recall, and F1-score across multiple benchmarks.	• Dynamic detection depends on test coverage quality, and incomplete coverage may miss certain logical failures. • The framework is currently validated only on Python, limiting cross-language generalizability.
[2]	LLM-Based Test-Driven Interactive Code Generation: User Study and Empirical Evaluation	IEEE Transactions on Software Engineering (TSE), 2024	• TICODER is a test-driven interactive workflow that incrementally clarifies user intent through LLM-generated tests and lightweight user feedback. • The system ranks and prunes candidate code suggestions based on user-approved tests using two variants: PASS/FAIL validation and explicit output specification.	• There is interactive overhead, and incorrect or noisy user feedback can prune correct solutions prematurely. • Effectiveness is contingent on the LLM's ability to generate high-quality tests.

[3]	A Federated Adaptive Large Language Model Fine-Tuning Framework for Software Development	IEEE Transactions on Services Computing, 2025	• The framework integrates LoRA-based parameter-efficient fine-tuning within a federated setting, where each organization locally trains lightweight adapter modules and shares only adapter to the central server. • A FedAvg-style is used to combine adapters based on local data size and an L2-norm constraint to manage gradient divergence	• Experimental results show degraded performance on unrelated datasets. • Although communication costs are reduced, the framework still requires multiple communication rounds, which may introduce latency in large-scale federations.
[4]	Towards Efficient Fine-Tuning of Language Models With Organizational Data for Automated Software Review	IEEE Transactions on Software Engineering 2024	• Proposes CodeMentor, a framework for few-shot learning that utilizes heuristic rules and weak supervision to construct datasets from internal organizational data • It employs Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning and integrates Reinforcement Learning from Human Feedback (RLHF) to incorporate domain expertise.	• The framework depends on the quality of organizational data, which may introduce domain bias and limit generalization across different coding lang. • RLHF requires computational resources and expert involvement, bottleneck scenario during scaling.
[5]	The Impact of Prompt Programming on Function-Level Code Generatio	IEEE Transactions on Software Engineering (TSE), 2025	• Empirical study to evaluate the impact of prompt programming techniques on function-level code generation. • CodePromptEval, a dataset of 7,072 prompts derived from 221 real-world Python functions, covering five prompt techniques: few-shot learning, chain-of-thought, persona, function signature, and package context, including all 32 possible combinations.	• The study is restricted to Python function-level tasks, limiting generalizability to other languages and higher-level program synthesis. • Rely on benchmark-driven evaluation, which may not fully capture real-world scenario. • Prompt techniques are evaluated in a fixed ordering, excluding effects of alternative prompt sequencing.

[6]	An Adaptive Framework Embedded With LLM for Knowledge Graph Construction	IEEE Transactions on Multimedia, Vol. 27, 2025	• KG construction happens in three stages: Information Extraction (IE) for open triple extraction, Additional Relation (AR) for enriching triples with semantic information, and Knowledge Graph Normalization (KGN) for schema-level alignment using vector similarity search. • Adaptive Prompt Generation (APG) module automatically generates, evaluates, and ranks prompts to guide LLM behavior across stages, enabling multi-domain KG construction from text, speech, and images.	• High computational overhead due to multi-stage LLM inference, embedding generation, and vector similarity matching. • The framework relies heavily on prompt engineering, which, is not fully autonomous and may still suffer from hallucination and semantic inconsistency.
[7]	Fight Fire With Fire: How Much Can We Trust ChatGPT on Source Code-Related Tasks?	IEEE Transactions on Software Engineering (TSE), Vol. 50, No. 12, 2024	• Evaluating self-verification capability of GPT for: code generation, code completion, and program repair. • Direct question, guiding question, and test report generation prompting is used assess whether ChatGPT can reliably validate its own outputs. • Effectiveness is measured using accuracy, precision, recall, and F1-score, with additional analysis of self-contradictory hallucinations.	• Results show that ChatGPT frequently overestimates the correctness of its own outputs, leading to low recall in direct self-verification. • While guiding questions and test reports improve recall, precision is reduced. • Generated test reports often contain incorrect explanations, especially for code generation and failed repairs.

[8]	FedALoRA: Adaptive Local LoRA Aggregation for Personalized Federated Learning in LLM	IEEE Internet of Things Journal, Vol. 12, No. 24, December 2025	- Proposes FedALoRA, a personalized federated learning framework that integrates LoRA-based PEFT with adaptive local aggregation to address severe non-IID data in federated LLM training. - FedALoRA uses an Adaptive LoRA Aggregation to merge global and local parameters via locally learned weights. This selective, element-wise fusion focused on higher transformer layers preserves domain expertise while capturing shared knowledge without increasing communication costs.	- FedALoRA introduces additional local optimization complexity due to adaptive weight learning in ALoRA modules. - The method requires careful tuning of the hyperparameter l, which controls the number of transformer layers involved in adaptive aggregation, and improper settings can degrade performance.
[9]	No Need to Lift a Finger Anymore? Assessing the Quality of Code Generation by ChatGPT	IEEE Transactions on Software Engineering (TSE), Vol. 50, No. 6, June 2024	- Evaluation framework for assessing ChatGPT-based code generation. The authors construct standardized prompts for 728 LeetCode algorithm problems and 54 CWE security scenarios across five programming languages. - Generated code is evaluated using automated judgment platforms (LeetCode for functional correctness) and static analysis tools (CodeQL for vulnerability detection). - A multi-round fixing process is designed, where execution or analysis feedback is iteratively fed back	- ChatGPT performs significantly better on problems published before 2021, indicating reliance on memorized patterns rather than reasoning. - The dialog-based fixing process has limited effectiveness, for logically complex problems . - Generated code often contains security vulnerabilities and non-deterministic behavior.

[10]	Knowledge Enhanced Program Repair for Data Science Code	Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (ICSE), 2025	• API KG Construction is done using a custom ontology and RDF triples to model API attributes and dependencies for seven data science libraries. • SPARQL is used for knowledge retrieval from on buggy code. • AST level Bug Knowledge Enrichment, which executes code node-by-node to localize runtime and assertion errors at the AST node level.	• DS-KG is version-specific and requires continuous updates to track evolving API documentation, making long-term maintenance costly and error-prone. • KG-based knowledge retrieval introduces higher latency than plain-text search
[11]	THINK: Tackling API Hallucinations in LLMs via Injecting Knowledge	IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2025	• THINK proposes a two-phase API hallucination mitigation framework consisting of Pre and Post execution Optimization. • Constructs an API knowledge database (Javadoc + validated code examples) and retrieves task-relevant APIs using task decomposition, similarity search, and LLM-based API labeling to guide code generation. • Error Classification into Seven API Error Types for correction	• It may introduce irrelevant API knowledge, occasionally degrading performance for tasks already solvable by strong LLMs. • Both retrieval results and ReAct-based repair depend on LLM generation, leading to non-deterministic behavior and reduced reproducibility across runs

[12]	EvoAPR: Enhancing Large Language Models for Automatic Program Repair with Genetic Algorithm and Dynamic LoRA	IEEE International Conference on Web Services (ICWS), 2025	• Genetic evolution of buggy hunks under tri-objective fitness: syntax compliance, bug-hunk masking rate, repair-location masking rate. • Dynamic LoRA swaps non-overlapping rank-r adapters per bug-index/project for on-demand, project-specific parameter retrieval. • Base + LoRA models generate candidate patches; token-level entropy scorer picks the least uncertain fix as the final patch.	• Genetic template evolution and dynamic LoRA fine-tuning introduce significant computation and implementation complexity compared to direct prompting-based APR methods. • The effectiveness of both stages relies on accurate bug localization and rich bug context.
[13]	CodeHalu: Investigating Code Hallucinations in LLMs via Execution Based Verification	Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025	• Executes LLM-generated code against test cases under time and memory constraints, recording execution failures, unexpected outputs, infinite loop behaviors to detect hallucination states. • Aggregates execution states across multiple LLMs and applies a two-stage heuristic induction process to classify hallucinations into four main types (Mapping, Naming, Resource, Logic) and eight subcategories.	• The study focuses exclusively on Python, and hallucination behaviors in other programming languages are not evaluated. • CodeHalu targets functional and logical correctness; identifying or mitigating security vulnerabilities is outside its scope.

ARCHITECTURE DIAGRAM OF THE PROPOSED SYSTEM

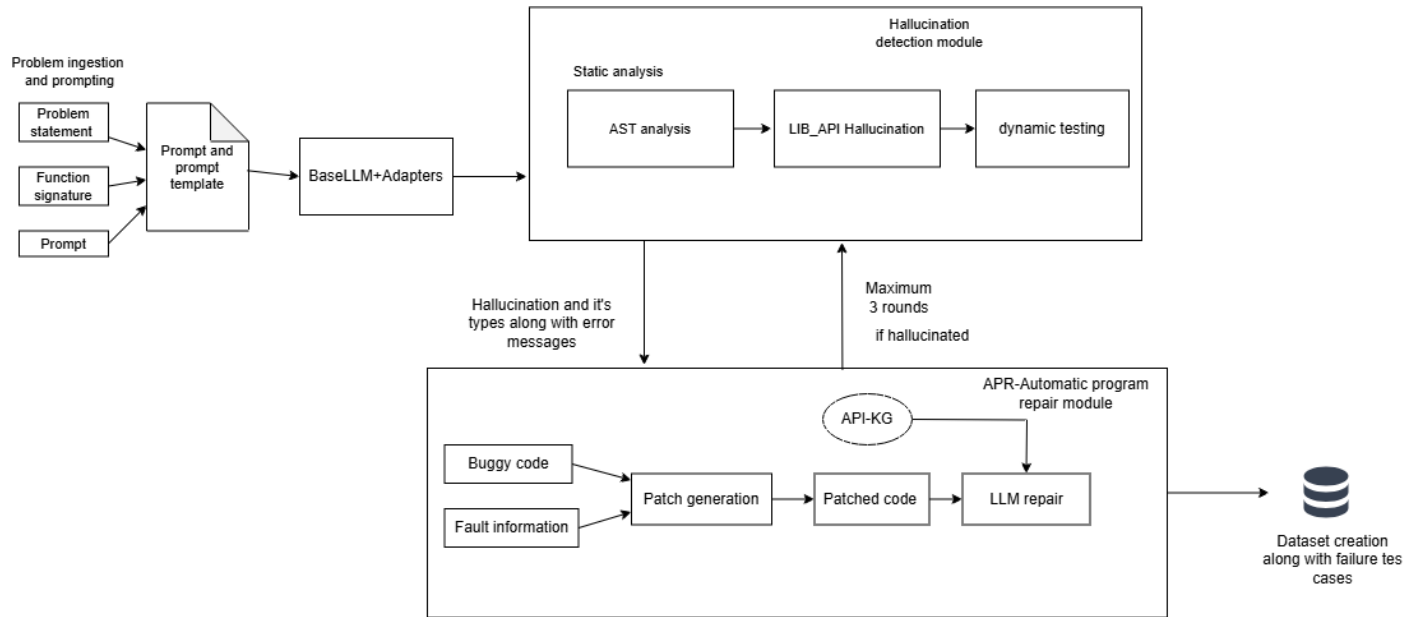


Fig1. Client side hallucination detection and mitigation framework

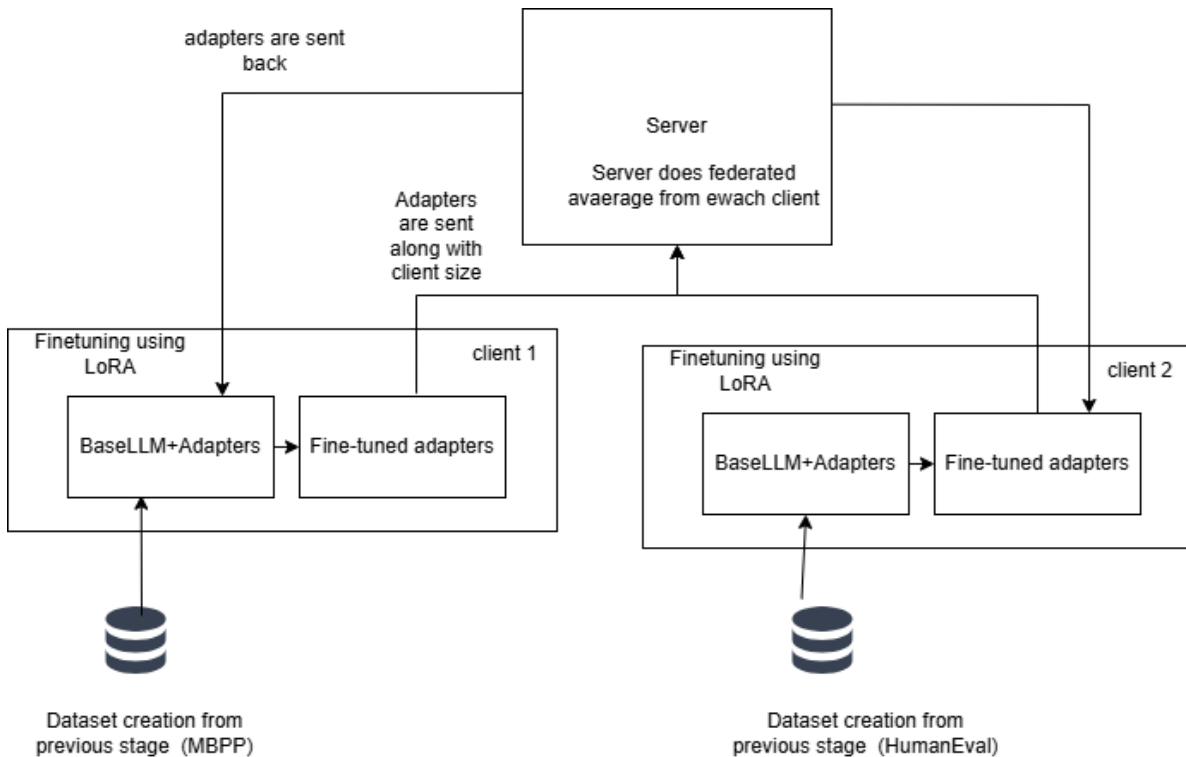


Fig 2. Federated learning + LoRA for fine-tuning

IDENTIFICATION OF PROJECT MODULES, TOOLS & PLATFORMS

S.NO	MODULES	SUB - MODULES	TOOLS & PLATFORMS
1.	Problem ingestion and code generation	- Dataset selection . - Client-wise dataset partitioning . - Single-shot prompting. - Pretrained LLM selection and loading. - Baseline inference using frozen LLM	Hugging Face Datasets, Transformers, Python, Pandas, NumPy, Google Colab.
2.	Base Model Selection and Configuration	- Static analysis using AST ,Library analysis. - Structured hallucination labeling.	Python ast module
3.	Automatic Program Repair (APR)	- Knowledge extraction. - Patch-based repair - Bounded repair attempts. - Ground-truth verification and hard hallucination labeling.	Numpy ,pandas, sklearn , scipy, statsmodel , Qwen2.5-Coder (Repair), Python.
4.	Fine tuning using LoRA	- Hallucination-aware dataset construction - Low-rank decomposition ($r = 8$). - Injection of LoRA modules into Query and Key projections. - Parameter isolation with frozen backbone. - Adapter checkpointing.	PyTorch, Hugging Face PEFT, Transformers.

5.	Federated Aggregation and Distribution	- Secure client-side adapters - Federated Averaging (FedAvg) with sample-size-based weighting. - Global adapter construction	Flower ,PyTorch, NumPy
6.	Evaluation and analysis	- Evaluation on MBPP, HumanEval, and DS-1000 benchmarks. - Functional correctness using Pass@1. - Hallucination detection metrics (Precision, Recall, F1). - Code similarity analysis using CodeBLEU. - Cross-client validation. - Before–after performance comparison.	CodeBLEU, Matplotlib, Seaborn, Scikit-learn

METHODS / MODELS

1. Problem ingestion and code generation

Input:

Benchmark code-generation datasets.

Output:

Initial LLM-generated code solutions (potentially containing hallucinations).

Methods:

- Dataset Loading:** Benchmark code-generation datasets including MBPP, HumanEval, and DS-1000 are loaded using the Hugging Face library.
- Client-wise Dataset Allocation:** Each federated client is assigned a distinct dataset or dataset subset to simulate heterogeneous behaviour.
- Problem Specification Construction:** For each task, structured inputs consisting of the problem statement, function signature, constraints is constructed.
- Single-Shot Prompting:** Code generation is performed using a single-shot prompting strategy, where the model receives the task specification without iterative refinement.
- Baseline Inference:** The frozen Qwen2.5-Coder-3B pretrained model is used to generate initial code outputs, which may contain hallucinations and serve as baseline for downstream analysis.

2. Hallucination detection

Input:

LLM-generated code obtained from the code generation module.

Output:

Detected hallucination types along with structured diagnostic information.

Methods:

- **AST Construction and Analysis:** Abstract Syntax Tree (AST) is constructed and analyzed to detect syntax errors, undefined variables, and invalid language constructs.
- **Library analysis:** We use ast to check if each methods, functions align with the library in static setup.
- **Dynamic Analysis:** The test cases are executed on the code to detect runtime errors and semantic hallucinations.
- **Hallucination Labeling:** Detected errors are categorized into different hallucination types, and the resulting information is recorded for dataset construction during fine-tuning.

3. Automatic program repair module

Input:

Buggy code, detected hallucination type, fault hints, and a domain knowledge graph suggestions.

Output:

Corrected code validated against ground-truth oracles or hard hallucination label.

Methods:

- **Knowledge Extraction:** library, class, method, and dependency information is extracted from dir, help, inspect .
- **Patch-Based Repair:** Applying localized patches to the hallucinated code from fault information from previous modules.
- **Fault-Guided Prompting:** The detected hallucination type, fault hints, and relevant knowledge graph entries are provided to LLM for code correction.
- **Bounded Repair Attempts :** program repair is performed with a bounded number of attempts to ensure controlled behavior.
- **GT Verification:** Each candidate is then verified with Hallucination detection module. If verification fails after bounded attempts, it is marked as hard hallucination.

4. Fine tuning using LoRA

Input:

Hallucination-aware code samples, including buggy code, diagnostic information, and repair outcomes, along with a frozen pre-trained LLM.

Output:

Client-specific LoRA adapters trained on hallucination aware datasets.

Methods:

- **Dataset Construction:** Construct structured dataset using problem description, buggy code, hallucination type, repairability flag and corrected code.
- **Low-Rank Decomposition:** Weight updates are parameterized using two low-rank matrices A and B, where the rank r is set to 8.
- **Attention Layer Injection:** LoRA modules are injected into the Query and Key projection matrices of each Transformer layer.
- **Parameter Isolation:** Original model weights remain frozen, and only LoRA parameters are made trainable.
- **Adapter Checkpointing:** Only trained LoRA adapter weights are saved and transmitted to the server, ensuring lightweight communication and privacy preservation.

5. Federated Aggregation and Distribution

Input:

Client-side LoRA adapter weights, along with sample size.

Output:

Global LoRA adapter distributed to all participating clients.

Methods:

- **Adapter Collection:** LoRA adapter weights are collected from all participating clients after local fine-tuning.
- **Federated Averaging:** Client-side adapters are aggregated using the Federated Averaging (FedAvg) algorithm, where clients with larger sample sizes contribute proportionally higher weights to the global adapter.
- **Global Adapter Construction:** A global LoRA adapter is constructed by averaging the weighted client adapters while keeping the base model parameters frozen.

- **Adapter Distribution:** The aggregated global LoRA adapter is redistributed to all clients and combined with the shared base model for subsequent inference or further training rounds.

6. Evaluation and analysis

Input:

Code generated by the base model and the federated LoRA-enhanced model across benchmark datasets.

Output:

Benchmark-wise evaluation.

Methods:

- **Functional Correctness (Pass@1):** The Pass@1 metric is used to measure functional correctness if code passes all unit tests on the first attempt.
- **Hallucination Detection Metrics:** Precision, Recall, and F1-score are computed based on oracle-validated hallucination labels to assess the accuracy of hallucination detection.
- **Code Similarity Analysis:** CodeBLEU is used as a secondary metric to measure syntactic and structural similarity between generated code and reference implementations.
- **Cross-Client Validation:** Cross-validation is performed across federated clients to evaluate generalization and robustness under heterogeneous data distributions.
- **Before-After Comparison:** Model performance is compared before and after LoRA fine-tuning to quantify improvements in correctness and hallucination reduction.

IMPLEMENTATION RESULTS

MODULE 1

Figure 3,4,5 shows the data loading for three dataset HumanEval, MBPP and DS1000 respectively.

Figure 6 shows the code snippet for code generation with QwenCoder template, which involves tokenization, decoding and code extraction.

```

from datasets import load_dataset
ds = load_dataset("openai/openai_humaneval")

*** README.md: 6.52k/? [00:00<00:00, 526kB/s]
openai_humaneval/test-00000-of-00001.par(...): 100% 83.9k/83.9k [00:00<00:00, 149kB/s]
Generating test split: 100% 164/164 [00:00<00:00, 3331.05 examples/s]

df_humaneval = pd.DataFrame(ds['test'])

print(df_humaneval.columns)

Index(['task_id', 'prompt', 'canonical_solution', 'test', 'entry_point'], dtype='object')

```

Fig 3

```

dataset = load_dataset("google-research-datasets/mbpp")
sanitized_dataset = load_dataset("google-research-datasets/mbpp", "sanitized")

mbpp_df = sanitized_dataset['train'].to_pandas()

def extract_signature(code: str):
    for line in code.splitlines():
        line = line.strip()
        if line.startswith("def "):
            return line
    return None
mbpp_df['function_signature'] = mbpp_df['code'].apply(extract_signature)

print(mbpp_df.columns)

Index(['source_file', 'task_id', 'prompt', 'code', 'test_imports', 'test_list',
      'function_signature'],
      dtype='object')

```

Fig 4

```

DS1000

ds1k = load_dataset("xlangai/DS-1000")
df_ds1k = ds1k['test'].to_pandas()
df_ds1k['prompt_2'] = df_ds1k['prompt'].str.split('A:', n=1).str[0].str.strip()

*** README.md: 100% 554/554 [00:00<00:00, 67.6kB/s]
testjsonl: 3.44M/? [00:00<00:00, 112MB/s]
Generating test split: 100% 1000/1000 [00:00<00:00, 17195.62 examples/s]

df_ds1k["task_id"] = [f"DS{str(i).zfill(3)}" for i in range(len(df_ds1k))]

print(df_ds1k.columns)

Index(['prompt', 'reference_code', 'metadata', 'code_context', 'prompt_2',
      'task_id'],
      dtype='object')

```

Fig 5

```

formatted_messages = construct_prompt(row['prompt'])
inputs = tokenizer.apply_chat_template(
    formatted_messages,
    add_generation_prompt=True,
    tokenize=True,
    return_dict=True,
    return_tensors="pt"
).to("cuda")

with torch.no_grad():
    outputs = model.generate(
        **inputs,
        max_new_tokens=512,
        do_sample=False,
        pad_token_id=tokenizer.eos_token_id
    )

gen_ids = outputs[0][len(inputs['input_ids'][0]):]
raw_response = tokenizer.decode(gen_ids, skip_special_tokens=True)
clean_code = extract_python_code(raw_response)

```

Fig 6

MODULE 2

Figure 7 shows hallucination distribution across all 3 datasets classified into syntax, indentation errors, structural hallucinations.

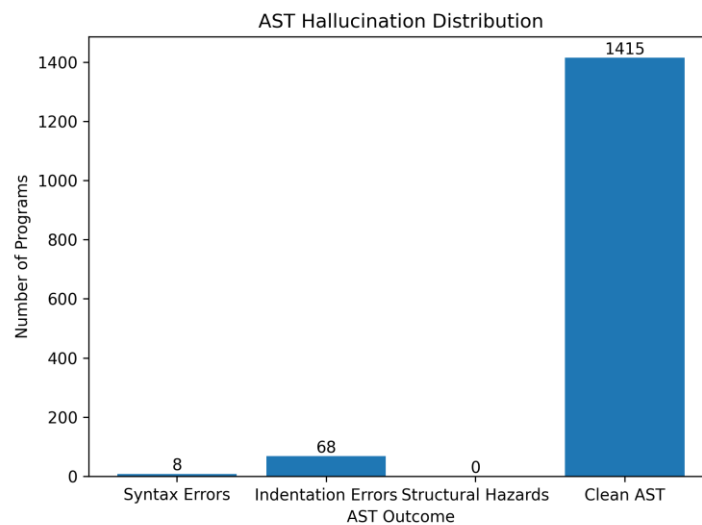


Fig 7

Figure 8 shows library or api hallucination across all dataset.

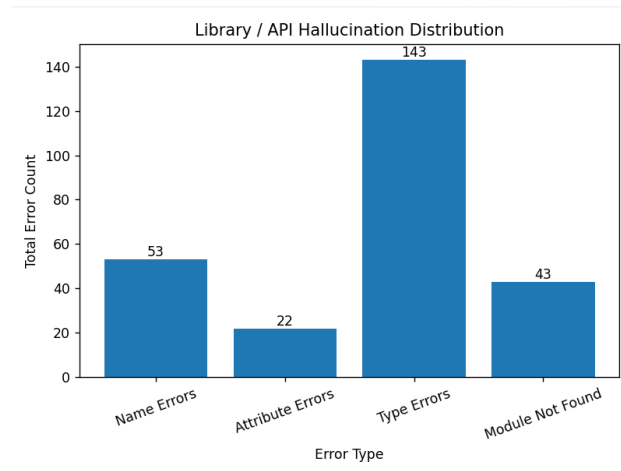


Fig 8

Figure 9 & 10 shows results after dynamic hallucination. Fig 9 describes the different type of errors encountered in a dataset. Fig 10 gives the overall hallucination pass and fail after dynamic analysis.

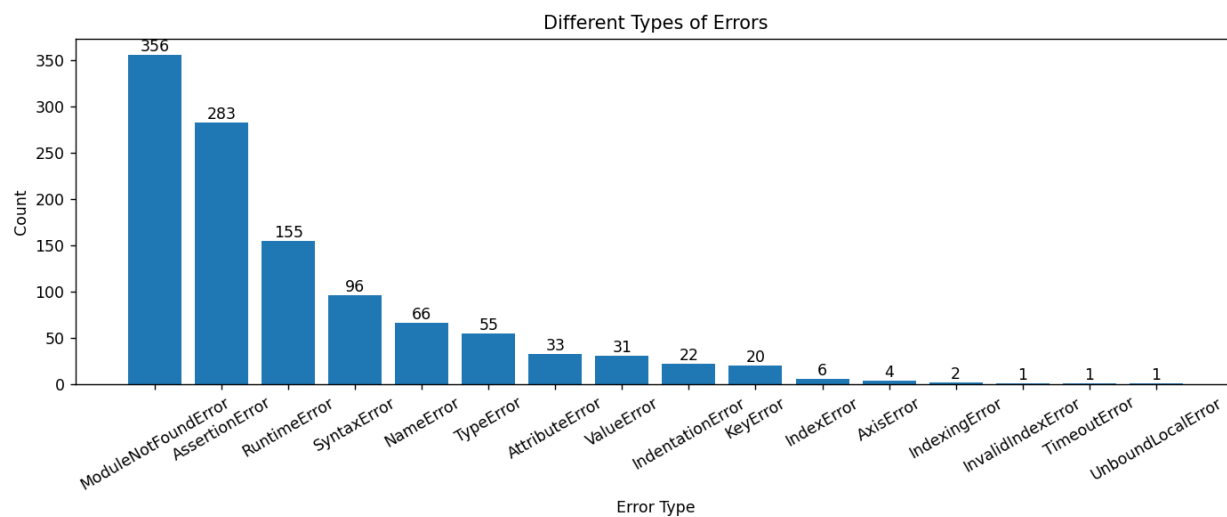


Fig 9

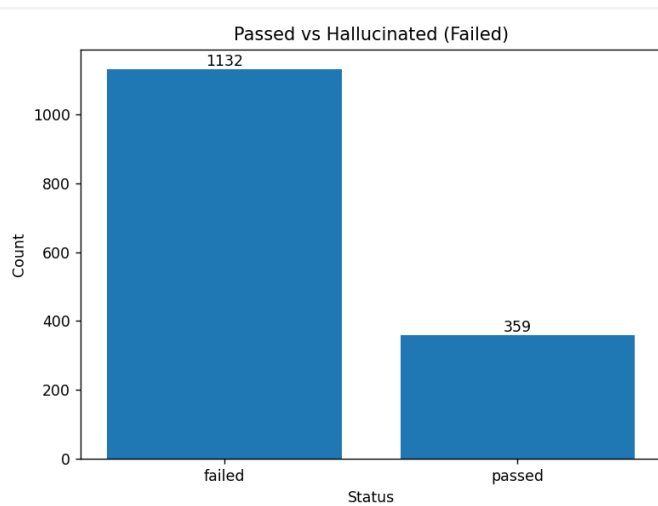


Fig 10

MODULE 3

Figure 11,12,13 shows an example from DS100 with task_id 42. We could see the patch generated around the error line and also the information to fix the error from fault_information.

```

df = pd.read_csv('test_ds1000.csv')
code = """
import pandas as pd
import numpy as np
df = test_input
result = df.iloc[1:].reset_index(drop=True).add_prefix('A_')
result = result.append(df.iloc[0].rename(columns=lambda x: f'B_{x}'))
"""

dataset_type = "ds1000"
task_id = "DS042"
row = task_row = df[df["task_id"] == "DS042"].iloc[0].to_dict()
result = do_pipeline(code,dataset_type,row,task_id)
print('Q RESULTS:')
print(result)

```

Fig 11

```

import pandas as pd
import numpy as np
df = test_input
result = df.iloc[1:].reset_index(drop=True).add_prefix('A_')
<<<< [ERROR START] (dynamic: AttributeError)
result = result.append(df.iloc[0].rename(columns=lambda x: f'B_{x}'))
[ERROR FINISH] (dynamic: AttributeError) >>>>

```

Fig 12

```

AST Info: {'type': None, 'value': 0, 'message': None}
Library Info: {'type': None, 'value': 0, 'libapi_details': []}
Dynamic Info: {'status': 'failed', 'error_type': 'AttributeError', 'error_message': "'DataFrame' object has no attribute 'append'",
{'dataset': 'ds1000', 'status': 'hallucinated', 'task_id': 'DS042', 'ast_info': '', 'lib_info': '', 'dynamic_info': '{"error_type":
KG Suggestions: [{'api': 'DataFrame.append', 'type': 'method', 'belongs_to': 'DataFrame', 'required_params': ['self', 'to_append'],
-----
Q RESULTS:
[[{'dataset': 'ds1000', 'status': 'hallucinated', 'task_id': 'DS042', 'ast_info': '', 'lib_info': '', 'dynamic_info': '{"error_type"

```

Figure 13

EXPERIMENTAL SETUP

- **Base LLM :**

- QwenCoder2.5-3B parameter is used as base model.
- Type: Causal Language Models
- Training Stage: Pretraining & Post-training
- Number of Parameters: 3.09B
- Number of Paramaters (Non-Embedding): 2.77B
- Number of Layers: 36
- Number of Attention Heads (GQA): 16 for Q and 2 for KV
- Context Length: Full 32,768 tokens
- 4-bit Quantized Model Loading: The model was loaded using 4-bit quantization with the bitsandbytes library.
- Deterministic Generation Settings: We configured do_sample = False and temperature = 1.0 to ensure deterministic and reproducible code generation across multiple runs.

- **Dataset:**

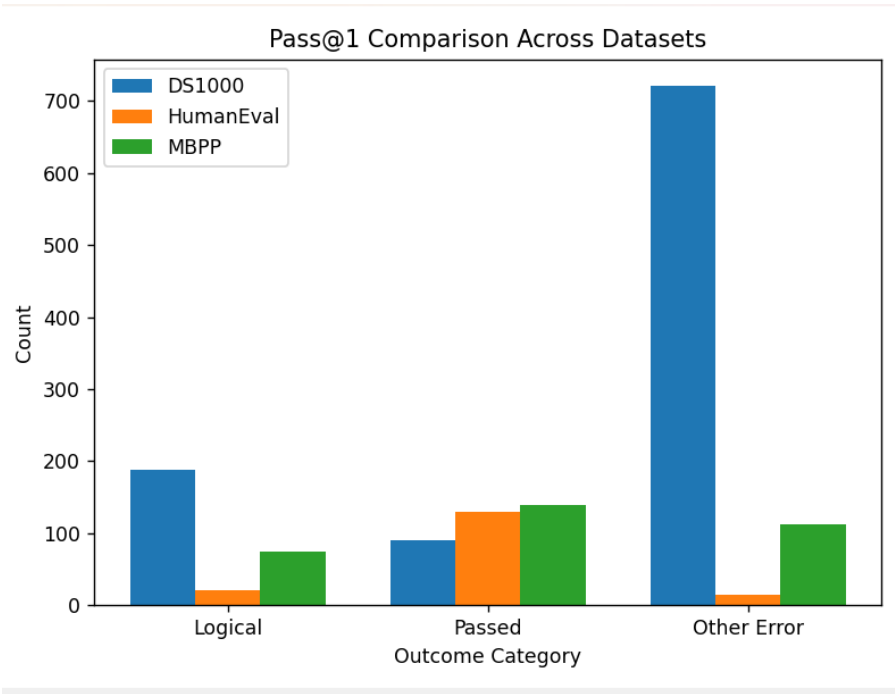
- **HumanEval:** Consist of 164 rows and 5 features.
- **MBPP:** Consist of 377 rows and 6 features. It is made up of most basic python programming.
- **DS1000:** Consist of 1000 data science problems in python.

- **Development environment:**

- **Google collab:** It is used for LLM inference, code generation ,building overall pipeline.It used NVIDIA T4 GPU with CUDA acceleration on runtime.
- **VS Code:** Local system is used to build static and dynamic modules which doesn't require GPU computation. Building modules and analysis is done on local from the generated dataset.

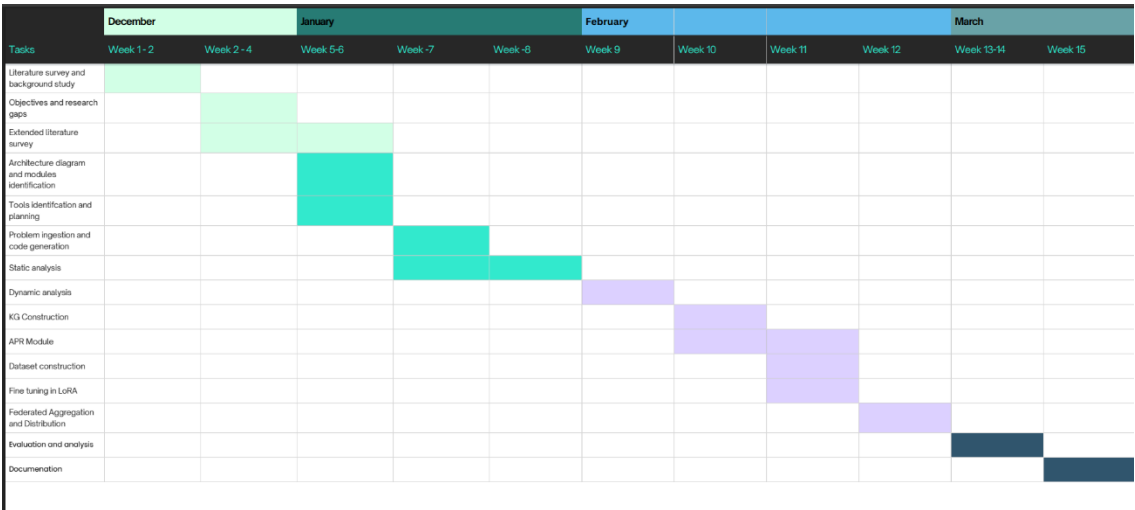
EVALUATION METRIC

Pass@1: The Pass@1 metric is used to measure functional correctness if code passes all unit tests on the first attempt.



Pass@1 comparison across dataset.

TIMELINE FOR COMPLETION OF MODULES



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